

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2023
Department of Materials Science and Engineering
MSE 3101
(Transport Phenomena in Materials)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Assume reasonable data if any missing.

Section A

1. (a) Write down the major assumptions used in the derivation of the Bernoulli equation and (12)
limitations on the use of the Bernoulli equation.
- (b) A large tank open to the atmosphere is filled with water to a height of 5 m from the outlet tap (12)
[Fig. 1(b)]. A tap near the bottom of the tank is now opened, and water flows out from the
smooth and rounded outlet. Determine the maximum water velocity at the outlet.

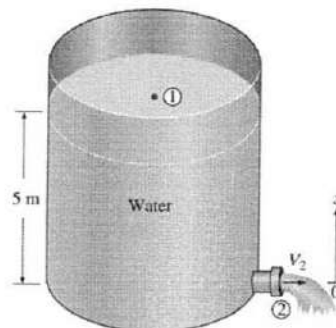


Fig. 1(b)

- (c) The viscosity of a fluid is to be measured by a viscometer constructed of two 42 cm long (11)
concentric cylinders. The outer diameter of the inner cylinder is 15 cm, and the gap between
the two cylinders is 0.16 cm. The inner cylinder is rotated at 350 rpm, and the torque is
measured to be 1.9 N.m. Determine the viscosity of the fluid.
2. (a) Explain the physical significance of thermal conductivity and thermal diffusivity. (10)
- (b) Starting with an energy balance on a rectangular volume element, derive one-dimensional (15)
transient heat conduction equation for a plane wall with constant thermal conductivity and no
heat generation.
- (c) Consider a 0.8 m high and 1.5 m wide glass window with a thickness of 8 mm. Determine (10)
the steady rate of heat transfer through this glass window and the temperature of its inner
surface for a day during which the room is maintained at 20°C while the temperature of the
outdoors is -10°C. Take the convection heat transfer coefficients on the inner and outer
surfaces of the window to be $h_1 = 10 \text{ W/m}^2 \cdot ^\circ\text{C}$ and $h_2 = 40 \text{ W/m}^2 \cdot ^\circ\text{C}$, which includes the
effects of radiation.
3. (a) What is critical radius of insulation? How is it defined for a cylindrical layer? (08)
- (b) Calculate the critical radius of insulation for asbestos [$k = 0.17 \text{ W/m} \cdot ^\circ\text{C}$] surrounding a pipe (12)
and exposed to room air at 20°C with $h = 3.0 \text{ W/m}^2 \cdot ^\circ\text{C}$. Calculate the heat loss from a 200°C,
5.0 cm diameter pipe when covered with the critical radius of insulation and without
insulation.
- (c) A hollow sphere of inside radius $r = a$ and outside radius $r = b$ is electrically heated at the
inner surface at a constant rate of $q_0 \text{ W/m}^2$. At the outer surface it dissipates heat by
convection into a fluid at temperature T_∞ with a heat transfer coefficient h . The thermal

conductivity k of the solid is constant. Develop expressions for the determination of the inner and outer surface temperatures T_1 and T_2 of the sphere. Calculate the inner and outer surface temperatures for $a = 3$ cm, $b = 5$ cm, $h = 400$ W/m²·°C, $T_\infty = 100^\circ\text{C}$, $k = 15$ W/m·°C, and $q_0 = 10^5$ W/m².

4. (a) Explain the Biot number for the lumped system analysis. (05)
- (b) Consider one dimensional, steady state heat flow along two stainless steel bars ($k = 20$ W/m·°C), each of diameter $D = 2$ cm, length $L = 3$ cm and pressed together with a pressure of 10 atm. The surface has a roughness of about 2.5 μm . An overall temperature difference of $\Delta T = 100^\circ\text{C}$ is applied across the bars. The interface temperature is about 90°C . The contact conductance of the interface is $h_c = 3000$ W/m²·°C. Calculate the heat flow rate along the bars. Calculate the temperature drop at the interface. (15)
- (c) The temperature of a gas stream is measured with a thermocouple. The junction may be approximated as a sphere of diameter $D = 1$ mm, $k = 25$ W/m·°C, $\rho = 8400$ kg/m³ and $c_p = 0.4$ kJ/kg·°C. The heat transfer coefficient between the junction and gas stream is $h = 560$ W/m²·°C. How long it take for the thermocouple to record 99 percent of the applied temperature difference? (15)

Section B

5. (a) Differentiate between the laminar flow and turbulent flow with diagram. (06)
- (b) With a neat sketch, explain laminar and turbulent velocity boundary layers that may form on a flat plate. (14)
- (c) Water flows at a velocity 1 m/s over a flat plate of length 0.6 m. The water temperature is 350K. In the laminar and turbulent regions, local convection coefficient are described by, $h_{lam}(x) = C_{lam}x^{-0.5}$; $h_{turb}(x) = C_{turb}x^{-0.2}$, where x has unit of m. At 350K, $C_{lam} = 477$ W/m^{1.5}K; $C_{turb} = 3600$ W/m^{1.8}K, $\rho = 974$ kg/m³, $\mu = 365 \times 10^{-6}$ N.s/m². Determine the average convection coefficient over the entire plate. (15)
6. (a) Baffles are installed in tube to increase the convection coefficient of the shell side fluid. Justify it with neat sketch. (05)
- (b) For an opaque surface, derive the net radiative flux equation, (15)

$$q''_{rad} = \epsilon \sigma T_s^4 - \alpha G.$$
- (c) "The counter flow heat exchanger is more efficient than the parallel flow heat exchanger." Explain the quotation with log mean temperature difference. (15)
7. (a) Prove that the solid angle of a radiative hemisphere is 2π sr. (10)
- (b) The spectral distribution of surface irradiation is shown in Figure 7(b). Calculate the total irradiation. (10)

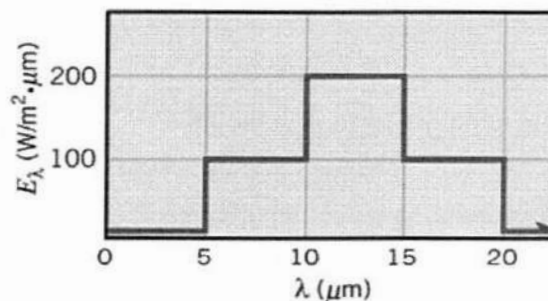


Figure 7(b)

- (c) Consider a large isothermal enclosure that is maintained at a uniform temperature of 2000K. (15)
- Calculate the emissive power of the radiation that emerges from a small aperture on the enclosure system.
 - What is the wavelength below which 20% of the emission is concentrated?
 - What is the maximum emissive power and wavelength at which this emission occurs? [Use Table 7(c)].

8. (a) Explain the term 'View Factor' for radiation transfer. (06)
- (b) Consider the diffusion of hydrogen in air at $T = 293\text{K}$. Calculate the species flux on both molar and mass bases, the concentration gradient $\frac{dc_A}{dx} = 1 \text{ Kmol}/(\text{m}^3 \cdot \text{m})$. Here, $D_{AB} = 0.41 \times 10^{-4} \text{ m}^2/\text{s}$ at 298K. (14)
- (c) Species A is evaporating from a flat surface into species B . Assume that the concentration profile for species A in the concentration boundary layer is of the form $C_A(y) = Dy^2 + Ey + F$, where D, E and F are constants at any x -location and y - is measured along a normal from the surface. Develop an expression for the mass transfer convection coefficient h_m in terms of these constants, the concentration of A in the free stream $C_{A,\infty}$ and the mass diffusivity D_{AB} . (15)

Table 7(c)

λT ($\mu\text{m} \cdot \text{K}$)	$F_{(0 \rightarrow \lambda)}$	$I_{\lambda,b}(\lambda, T)/\sigma T^5$ ($\mu\text{m} \cdot \text{K} \cdot \text{sr}$) ⁻¹	$\frac{I_{\lambda,b}(\lambda, T)}{I_{\lambda,b}(\lambda_{\text{max}}, T)}$
200	0.000000	0.375034×10^{-27}	0.000000
400	0.000000	0.490335×10^{-13}	0.000000
600	0.000000	0.104046×10^{-8}	0.000014
800	0.000016	0.991126×10^{-7}	0.001372
1,000	0.000321	0.118505×10^{-5}	0.016406
1,200	0.002134	0.523927×10^{-5}	0.072534
1,400	0.007790	0.134411×10^{-4}	0.186082
1,600	0.019718	0.249130	0.344904
1,800	0.039341	0.375568	0.519949
2,000	0.066728	0.493432	0.683123
2,200	0.100888	0.589649×10^{-4}	0.816329
2,400	0.140256	0.658866	0.912155
2,600	0.183120	0.701292	0.970891
2,800	0.227897	0.720239	0.997123
2,898	0.250108	0.722318×10^{-4}	1.000000

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2023
Department of Materials Science and Engineering

MSE 3103
(Materials Manufacturing Processes)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Assume reasonable data if any missing.

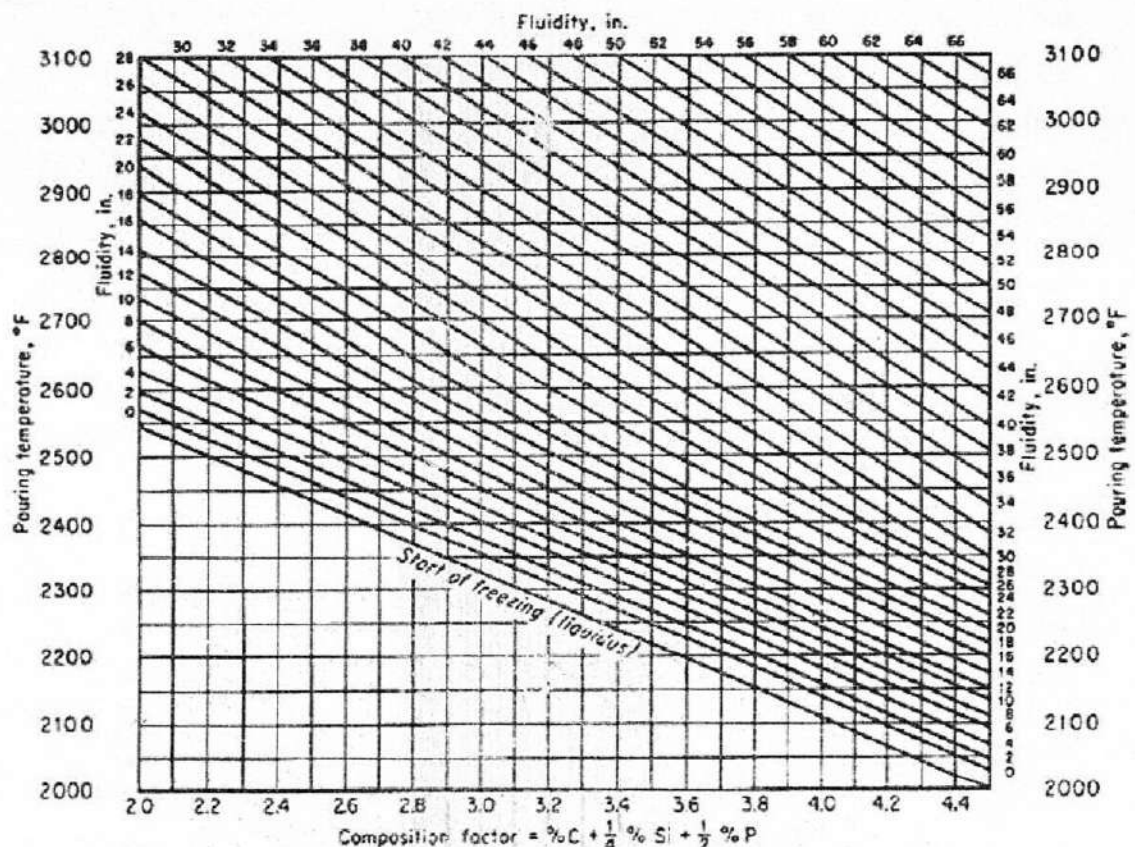
Section A

1. (a) Why casting process is considered as a major manufacturing process? Explain it mentioning (08)
with advantages.
- (b) Differentiate between expandable and permanent mold process. Which one will you choose (07)
for complex shape casting?
- (c) Describe the general characteristics of greensand molding during molding, casting and storage (10)
stages.
- (d) With the help of schematic diagram explain the effect of clay and water on the maximum green (10)
compressive strength of clay-sand-water mixtures.
2. (a) What are the resistances to heat flow across a casting freezing in a mold? Explain with (11)
temperature profile curve.
- (b) Describe the classical theory of nucleation for a spherical particle with the free energy change (12)
curve associated the radius r of a solid nucleus.
- (c) Explain the freezing of impure metals and alloy due to undercooling with schematic (12)
illustration of solid/liquid interface growth mode variation.
3. (a) What are the criteria of obtaining an ingot entirely of equiaxed crystals? Also, describe the (12)
mechanism of equiaxed crystals formation.
- (b) State the seven rules of feeding design. (07)
- (c) A foundry received an initial order of 5000 pieces of Al cylinder casting (400 mm length, 300 (08)
mm outer dia and 260 mm inner dia) with a possibility of repeat order. It was decided to
produce a master pattern of aluminium for casting multiple working pattern. Using reasonable
allowances, determine the dimensions of the master pattern & its core print. Density of Al =
 2.7 g/cm^3 and core sand = 1.6 g/cm^3 . Compressive strength of core sand = 2.0 kg/cm^2 , chaplet
= 2.2 kg/cm^2 & factor of safety = 4.
- (d) Design suitable feeding systems for the casting of above prototype (Question 3(c)) 08
4. (a) State the functions of a good gating system. (08)
- (b) Explain major casting defects and their reasons. (12)
- (c) Design a suitable gating system for a cylindrical pipe (150 mm outer diameter, 15 mm (15)
thickness, 200 mm length) made using grey iron of composition 3.0% C, 1.8% Si, 0.3% P and
density 7.0 g/cc . The moulding system to be used is green sand and pouring temperature is
 2450°F . Assume reasonable value for any missing data. List all basis and assumptions you
made during the design. Empirical formulae for pouring time (in seconds) calculations of thin
walled grey iron casting of thickness 8-15 mm and weight less than 450 kg is $t = 2.2\sqrt{w}$,
where, w is the weight of the liquid metal poured. [Use Fluidity chart for Question 4(c)].

using gunmetal listing all assumptions, if any, you made during the design. (Assume reasonable value if any missing data).
Given: Gunmetal density = 8.7 g/cm^3 , volume expansion coefficient = 7.0%. Pouring time, $t = 1.8 (\delta w)^{1/3} \text{ sec}$. (δ = avg. thickness in mm).

Section B

5. (a) Define hot working. Write down the advantages and disadvantages of hot working. (11)
- (b) Describe the sheet metal forming process. What are the defects and remedy of sheet metal forming? (12)
- (c) A 300 mm wide strip 25 mm thick is fed through a rolling mill with two powered rolls each of radius = 230 mm. The work thickness is to be reduced to 22 mm in one pass at a roll speed of 45 rev/min. The work material has a flow curve defined by $k = 270$ MPa and $n = 0.13$, and the coefficient of friction is assumed to be 0.11. Determine if the friction is sufficient to permit the rolling operation to be accomplished. If so, calculate the roll force, torque and horsepower. (12)
6. (a) Write down the different cutting conditions for roughing cut and finishing cut. (10)
- (b) Distinguish between shaping and planning operations. (12)
- (c) In a machining operation that approximates orthogonal cutting, the cutting tool has a rake angle of 10 degree. The chip thickness before the cut $t_0 = 0.50$ mm and the chip thickness after the cut $t_c = 1.125$ mm. Calculate the shear plane angle and the shear strain in the operation. (13)
7. (a) State the factors affecting the polymer crystallinity. (12)
- (b) Distinguish between vacuum thermoforming and pressure thermoforming processes. (10)
- (c) As a materials engineer, choose a suitable type of glass for automobile windshields with proper justification. (13)
8. (a) Briefly describe the steps required for a powder metallurgy process. (12)
- (b) Explain how does the sintering operation reduce porosity and improve mechanical strength of ceramic powder products. (11)
- (c) Choose an additive manufacturing process for fabricating polymer products considering the energy requirement. (12)



Question 4(c): Fluidity chart

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2023
Department of Materials Science and Engineering

MSE 3105
(Welding and Materials Joining Processes)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

1. (a) What are the differences between soldering and brazing? (10)
(b) Classify different types of mechanical and metallurgical joining process with brief description. (10)
(c) Suppose your welding company is considering between two types of dip brazing station, namely, molten metal bath and molten chemical bath. Differentiate between each brazing setup and evaluate their advantages so that they can decide. (15)
2. (a) With appropriate diagrams, describe the types of weld joints. (10)
(b) Name and explain the essential components of a weld symbol, and provide suitable examples. (10)
(c) Discuss the modes of metal transfer in GMAW with necessary figures. Choose which mode of metal transfer is suitable for out of position welding and analyze why? (15)
3. (a) Describe unique advantages of Diffusion welding. (10)
(b) Differentiate the prominent features between DCEN and DCEP power supply mode in TIG welding. (12)
(c) Imagine, you are given with one Aluminium and one steel sample (incompatible metals). Suggest the friction welding method that can be used to join these two dissimilar metals with proper reasoning. (13)
4. (a) List out the advantages and disadvantages of EBW and LBW high energy beam welding. (10)
(b) Explain direct drive and inertia drive friction welding mechanism. (12)
(c) Choose the appropriate process between Explosion welding and Roll welding that can be used to weld heavy section over a large surface area and justify your choice. (13)

Section B

5. (a) What is the melting efficiency? What is the effect of the increasing power density of the welding heat source? (08)
(b) Sketch a weld pool microstructure, label different regions and explain any three nucleation mechanism with diagrams. (15)
(c) Derive the mathematical relationship between the growth rate R and the welding speed V , both at the centerline and at the fusion line. (12)
6. (a) What is liquation cracking? (05)

- (b) You are given an Al-6.3% Cu alloy for welding, where the Cu concentration C_2 is outside the solubility limit of the α phase ($C_2 > C_{SM}$). Explain the liquation mechanism of the C_2 alloy, which consists of α matrix and A_xB_y particles in the as-cast condition. Using the phase diagram of Figure 6(b), explain what happens to the base metal and PMZ regions. Also, draw the grain microstructures before and after welding. (15)

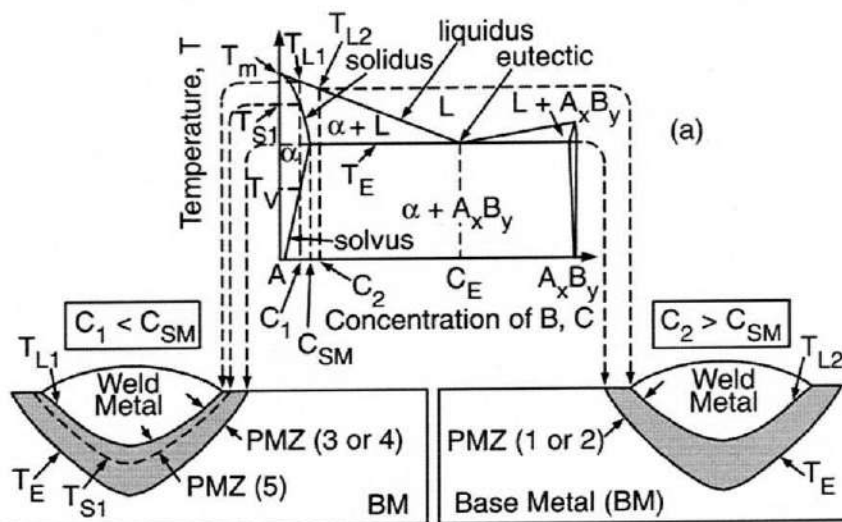


Figure 6(b)

- (c) Suppose you are the senior quality control engineer in a welding company. The company always uses stabilized austenitic stainless steel 321 and 347, to avoid any kind of weld decay. But they are still observing some corrosion attack in the narrow region immediately adjacent to the weld metal. Can you identify why this corrosion attack may be happening? Can you propose two or three solutions that they can try to remedy this? (15)
7. (a) What is the reason behind the weld decay in austenitic stainless steels? Describe two remedies for avoiding such decay. (10)
- (b) "High-carbon martensite can form in the heat affected zone of an as-cast low-carbon steel during welding" – explain how the martensite can form in this case using Fe-C phase diagram. (10)
- (c) You are in charge of supervising all the welding process in a plant. One of your technicians reported to you that the welding of a precipitation-hard Aluminium alloy sees hardness loss in the heat affected zone and showed you the measured hardness plot below [Figure 7(c)]. He does not know why. (15)
- (i) As the lead engineer, explain to him why the hardness drops during welding, by analyzing what happens at locations 1–4 with thermal cycles and phase diagram.
- (ii) Propose a solution to increase the hardness using post-weld artificial aging and explain how it will affect the locations 1–4. Draw the final expected hardness profile vs. location.

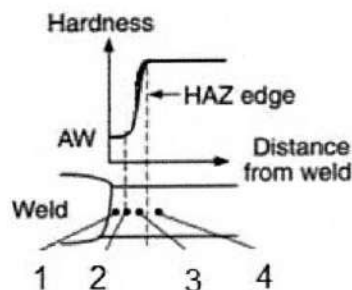


Figure 7(c)

8. (a) Describe the following terms: (08)
- (i) Stress corrosion cracking (ii) Fatigue.
- (b) Explain the formation of distortion in weld metal with remedies. (12)
- (c) Assume you are assigned as a NDT expert to investigate the surface and slightly subsurface defects. Defend your choice of NDT technique for this kind of investigation. (15)

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2023
Department of Materials Science and Engineering

MSE 3107
(Physical Metallurgy of Materials)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Assume reasonable data if any missing.

Section A

1. (a) Explain the recrystallization mechanism of a cold worked sample with necessary diagram. (11)
(b) Explain the allotropic transformations of iron at 1539°C, 1400°C and 910°C detailing the crystal structures and magnetic transitions associated with each phase. (12)
(c) Draw an I-T diagram and sketch the cooling curves to obtain (i) 100% bainite (ii) 30% pearlite, 30% bainite and the remaining martensite (iii) 50% fine pearlite and 50% martensite. (12)
2. (a) 'Annealing can never be a final heat treatment process.' – Justify the statement. (10)
(b) In case of hypereutectoid steel, the temperature chosen for normalizing exceeds the upper critical temperature, whereas for annealing, it surpasses the lower critical temperature. Conversely, for hypoeutectoid steel, both annealing and normalizing are conducted above the upper critical temperature. Illustrate the rationale behind this practice. (13)
(c) Design the annealing and normalizing schedules for 4340 steel bars having 40 mm diameter. (12)
The chemical composition of 4340 grade steel bars is given below:

AISI-SAE Number	%C	%Mn	%Ni	%Si	%Mo
4340	0.38-0.43	0.60-0.80	1.65-2.00	0.15-0.30	0.20-0.30

3. (a) Explain the effects of austenitization temperature on the incubation period of austenite formation and its growth in a typical eutectoid steel. (10)
(b) Discuss the recommended heat treatment procedures based on different CE value ranges to ensure safe operation and minimize the risk of cracking. (12)
(c) Determine the CE value of the following steels and comments on heating rate and sensitivity. (13)

AISI-SAE Number	%C	%Mn	%Si	%Ni	%Cr	%Mo	%Y
4340	0.40	0.40	0.30	2.00	0.70	0.20	-
8620	0.20	0.70	0.15	0.40	0.60	-	0.20

4. (a) Illustrate how the tempering temperature affects the mechanical properties of steel. (10)
(b) Summarize the advantages of oil medium over water in the quenching process, particularly in terms of its ability to handle intricate shapes and high hardenability alloys. (10)
(c) You are a materials engineer at a company specializing in the manufacturing of high performance steel components for the automotive industry. Your team is working on two projects: (15)
(i) Developing a new steel alloy for heavy-duty truck axles.
(ii) Producing a set of gears for high speed transmissions.

Demonstrate the factors affecting the hardenability of steel alloys for both projects. Include relevant diagrams to support your analysis.

Section B

5. (a) Summarize the key points of the surface hardening process, including its techniques, benefits, and applications. Highlight the main reasons for its widespread use in engineering. (10)
- (b) Discuss the pack carburizing process on a low C steel sample mentioning its microstructural changes. (10)
- (c) A low steel has 0.15% C with Cr and Mo which needs to be carburized by pack/gas carburizing method at 900°C. Calculate the carburizing time for the low alloy steel with 1% Cr / 1% Mo separately. Use Figure 5(c)_1 and Figure 5(c)_2 to conduct the calculation. (15)
6. (a) Discuss the purposes of post oxidation treatment of a nitrided sample. (08)
- (b) Illustrate the microstructural changes that occur in steel during cyaniding process. (12)
- (c) A laser heating system of 1500 W with convergent lens is being used for one second in transformation hardening of an alloy steel surface. If the focal length, spot size and wavelength of the laser beam are, respective, 300 mm, 1.5 mm and 10 mm, calculate the size of the laser beam, M factor value. Consider beam scan rate of 400 mm/min. Give your comment for this LTH. (15)
7. (a) Compare the effects of carbon content and graphite morphology on the mechanical properties and performance of gray cast iron and ductile iron. (10)
- (b) “Aging time and temperature have significant impact on the yield strength of an Al-4% Cu alloy.”— Justify the statement with appropriate figure. (10)
- (c) Discuss the common annealing treatments required for Ti alloys to gain fracture toughness, ductility, and creep resistance at room temperature. (15)
8. (a) “Dual phase steels offer an outstanding combination of strength and drawability through a special heat treatment process.”— Justify the statement with diagram. (10)
- (b) Explain the effects of microalloying elements such as niobium, vanadium and titanium in refining the grain microstructure and how do these elements facilitate precipitation hardening in microalloyed steels? (11)
- (c) You are a material engineer tasked to devise a strategy to obtain high toughness and hardness from the TMT alloy. Explain your approach using necessary figures to illustrate key concepts. (14)

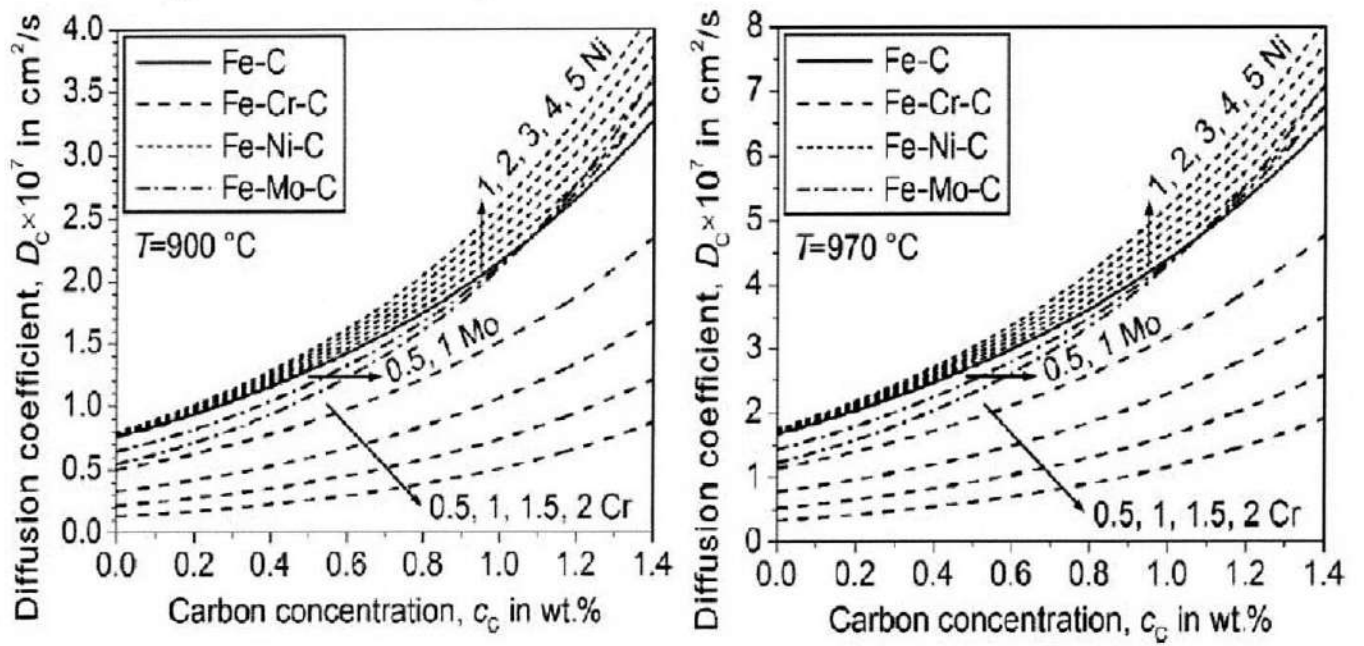


Figure 5(c)_1: Diffusivity prediction of the content of Cr, Mo, and Ni.

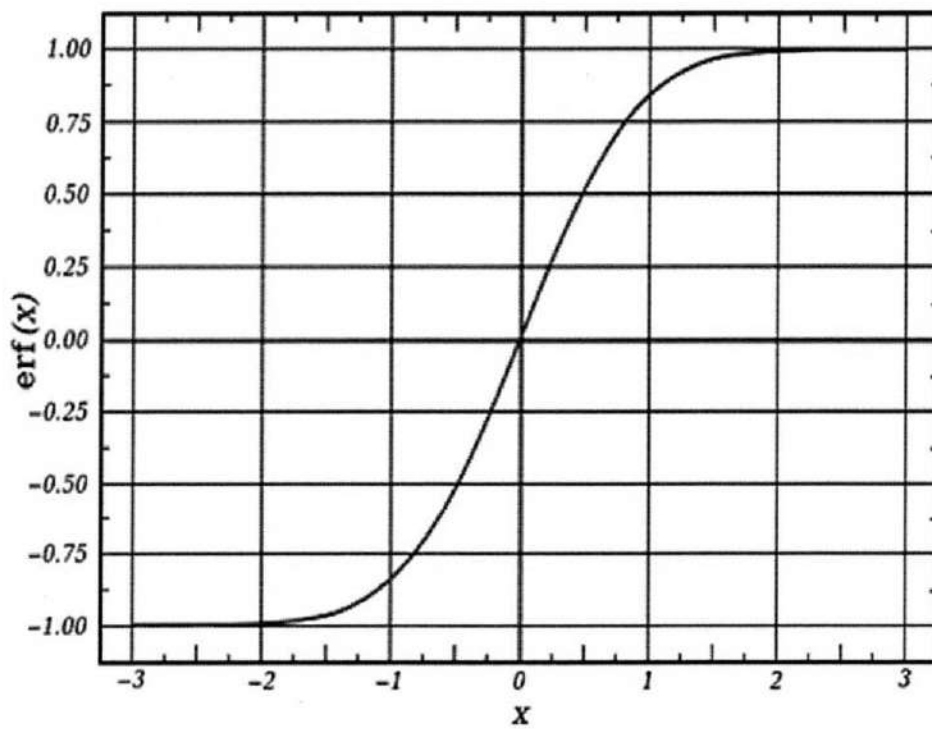


Figure 5(c)_2: Error function curve in term of distance.

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2023
Department of Materials Science and Engineering

Hum 3127
(Government and Sociology)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

1. (a) Define politics. People do say that politics is a dirty game, is it true? Discuss. (10)
(b) What is state and government? Distinguish between the compulsory and optional functions of a modern welfare state. (15)
(c) What is socialism? Describe the future of socialism. (10)
2. (a) Explain the origin and development of the state according to the Devine origin theory. (10)
(b) What is fascism? Give an account of Mussolini's fascism in brief. Discuss the problems that fascism in Italy could not solve. (15)
(c) What are the methods of securing the independence of judiciary? (10)
3. (a) What are the benefits of good citizenship? Do you think you are a good citizen? Why or why not? (15)
(b) What is Marxism? Discuss Marxism and its contribution to the world politics. (10)
(c) What is Legislature? Describe the functions of Legislature. (10)
4. (a) What is the relationship between the presidential and parliamentary form of government? Is a presidential system more democratic than a parliamentary system? Why or why not? (20)
(b) What is constitution? Do you think that the constitution of Bangladesh is a good constitution in the world and why? (15)

Section B

5. (a) How does sociological insights differ from common sense explanations of social behavior? (12)
(b) Identify the origin of sociology to historical social changes. (10)
(c) Find the advantages of sociological thinking for our personal growth. (13)
6. (a) Explain the influence of context and norms in social interaction. (10)
(b) How does social conformity affect us? (10)
(c) Compare and contrast deterrence theory and Travis Hirschi's control theory to explain juvenile delinquency. (15)
7. (a) Explain socio-cultural pre-requisites of Industrialization. (10)
(b) Why cultural lag has become a problem in modern society? (10)
(c) What is modernization? Identify the sources of social change due to modernization. (15)

8. (a) To what extent is gender a social construction? (10)
- (b) What is Xeno-centrism? Analyze the biological and cultural perspective of human behavior. (10)
- (c) Illustrate the ecological crisis evolving in modern times. (15)